

Plant Water Management Experiments 5 & 6 on ISS: Hydroponics and Ebb and Flow

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Some Benefits of E/F and Hydroponics as option for plant watering in space

- Clean
- Cleanable
- No substrate
- Good growth
- Semi-passive control
- Automatable
- ...

Microgravity Fluid Physics Challenges

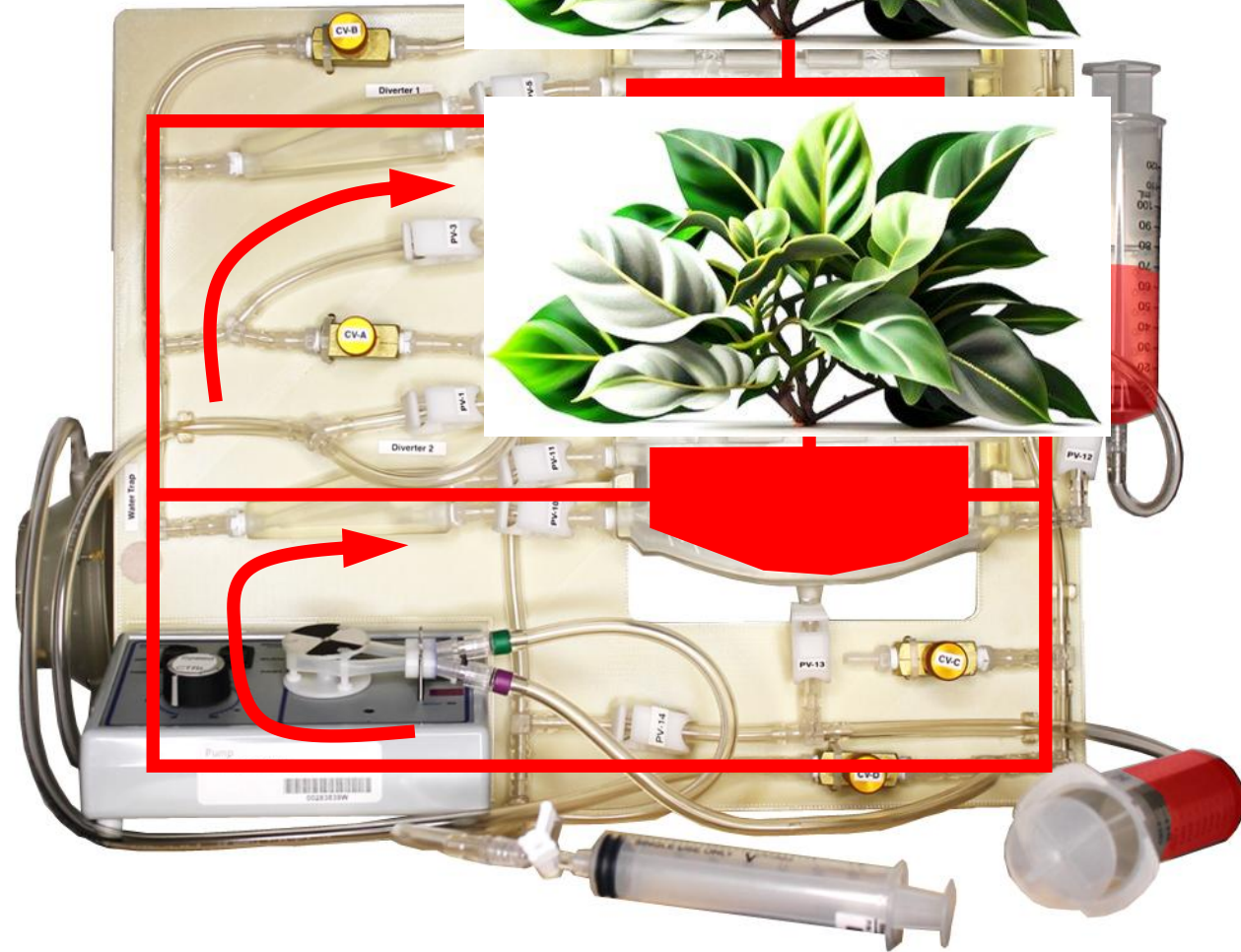
- Capillary fluidics over large length scales
- Unsteady developing transitional flows
- Instability of rogue bubbles, films, rivulets, jets, geysers, and drops
- Poor, variable wetting, hysteresis, and pinning
- Complex free surface geometries
- Myriad metastable states
- Stability of parallel inertial-capillary channels
- Roots may not be that predictable
- Need for robust long duration solutions, insensitive to ‘excursions’

The μ -g Fluid Physics of the PWM...

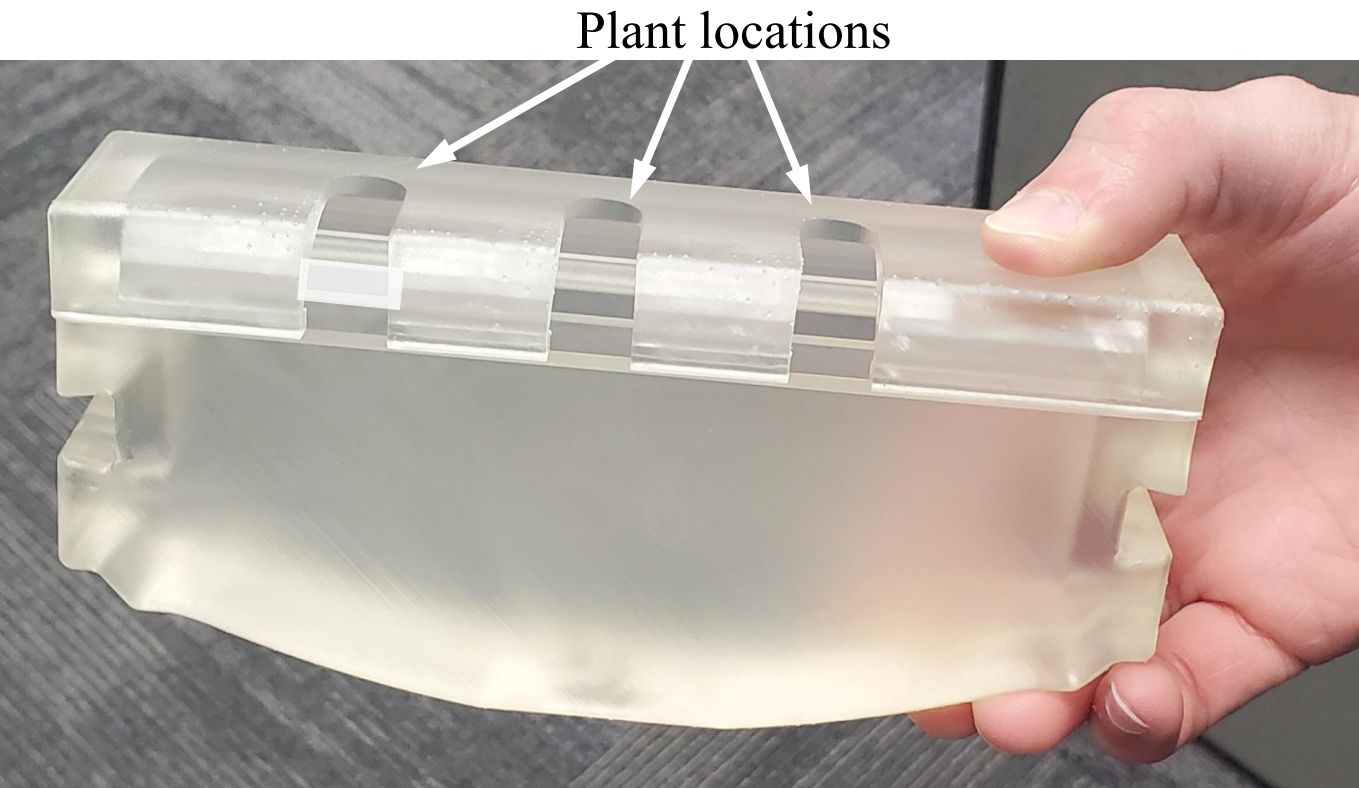
PWM-5

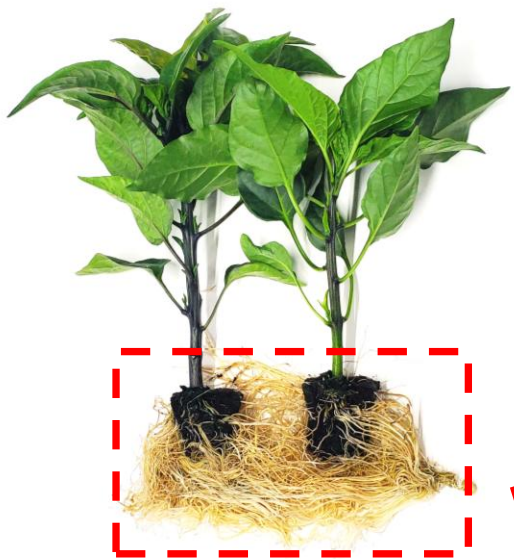


PWM-6

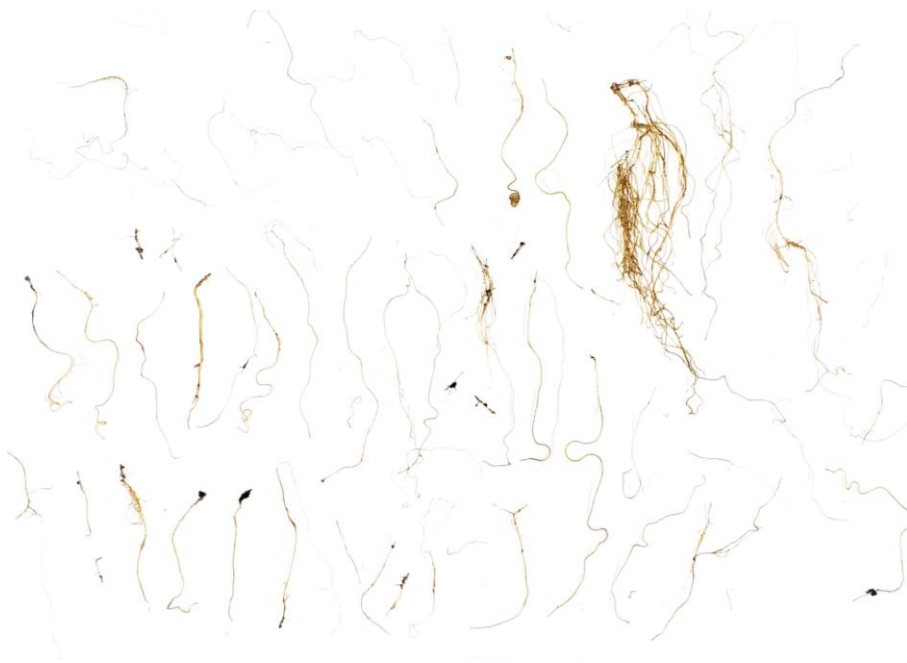


PWM 5 & 6 root model development





45d Peppers
in wedge channel



110 roots
 $1'' < L < 10''$
Even distribution



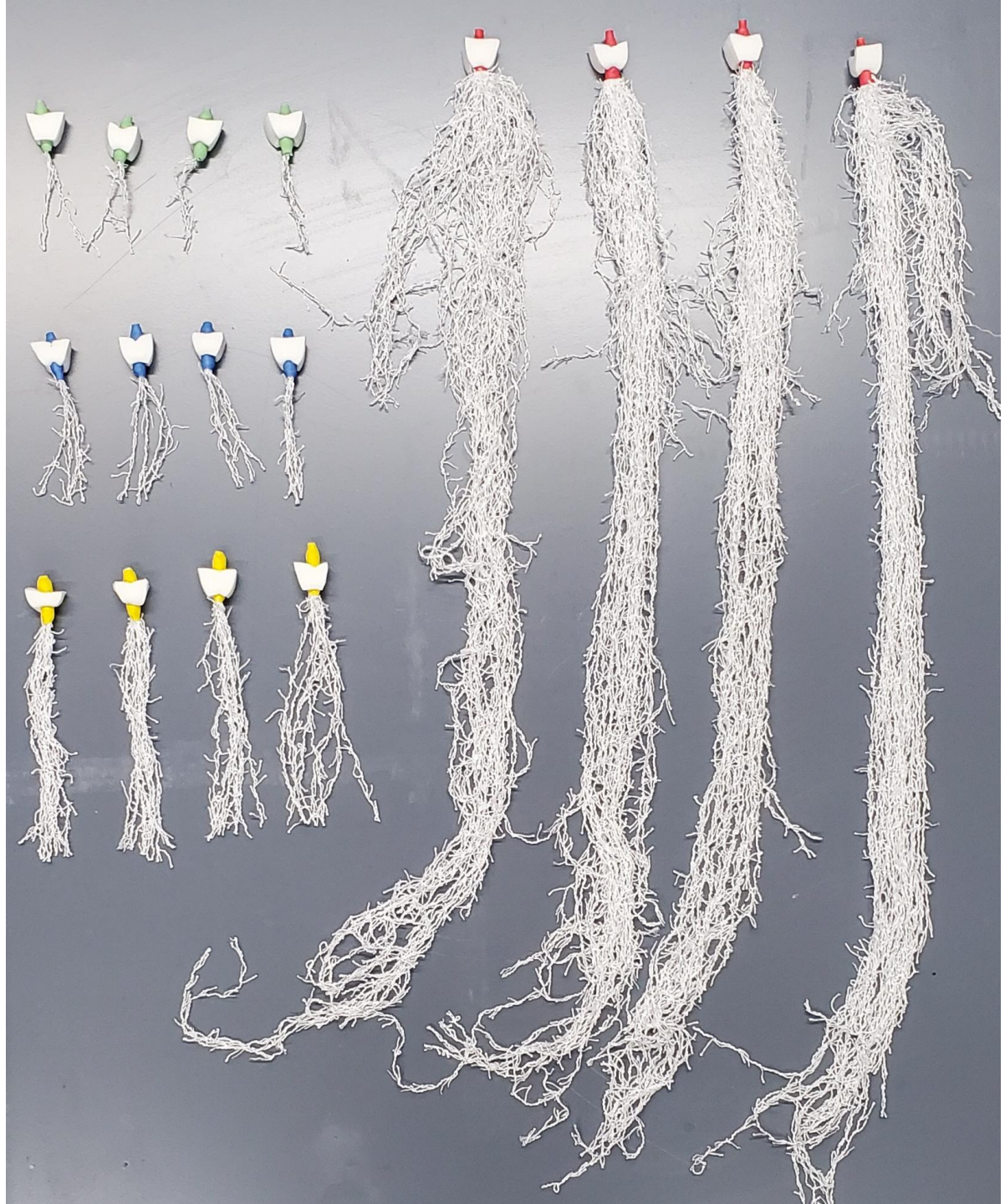
Flight Root Models

#1: Sparse: 2 Roots, 2"

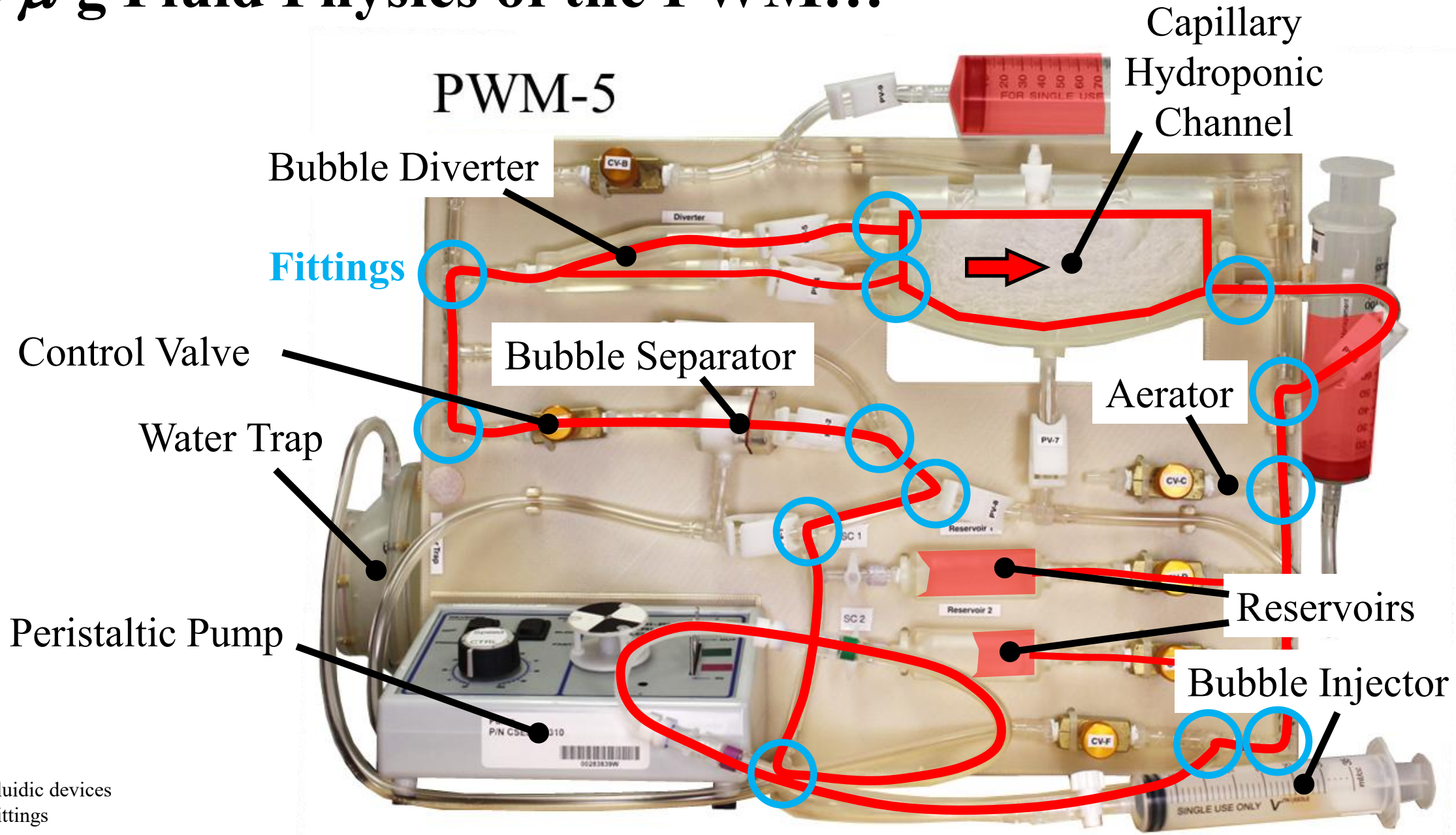
#2: Small: 4 Roots, 2.5"

#3: Medium: 8 Roots, 5"

#4: packed: 30 Roots, 6" & 24"



The μ -g Fluid Physics of the PWM...



- 10 Capillary fluidic devices
- 13 Capillary fittings

The μ -g Fluid Physics of the PWM...

- Jenson et al., *Int. J. Multiphase Flow*, **65**, 2014.
- Weislogel et al., US Pat. No. 9,962,024, 2018.

PWM-5

Bubble Diverter

- Weislogel et al., Gas Diverter, US Pat. App., 2024.

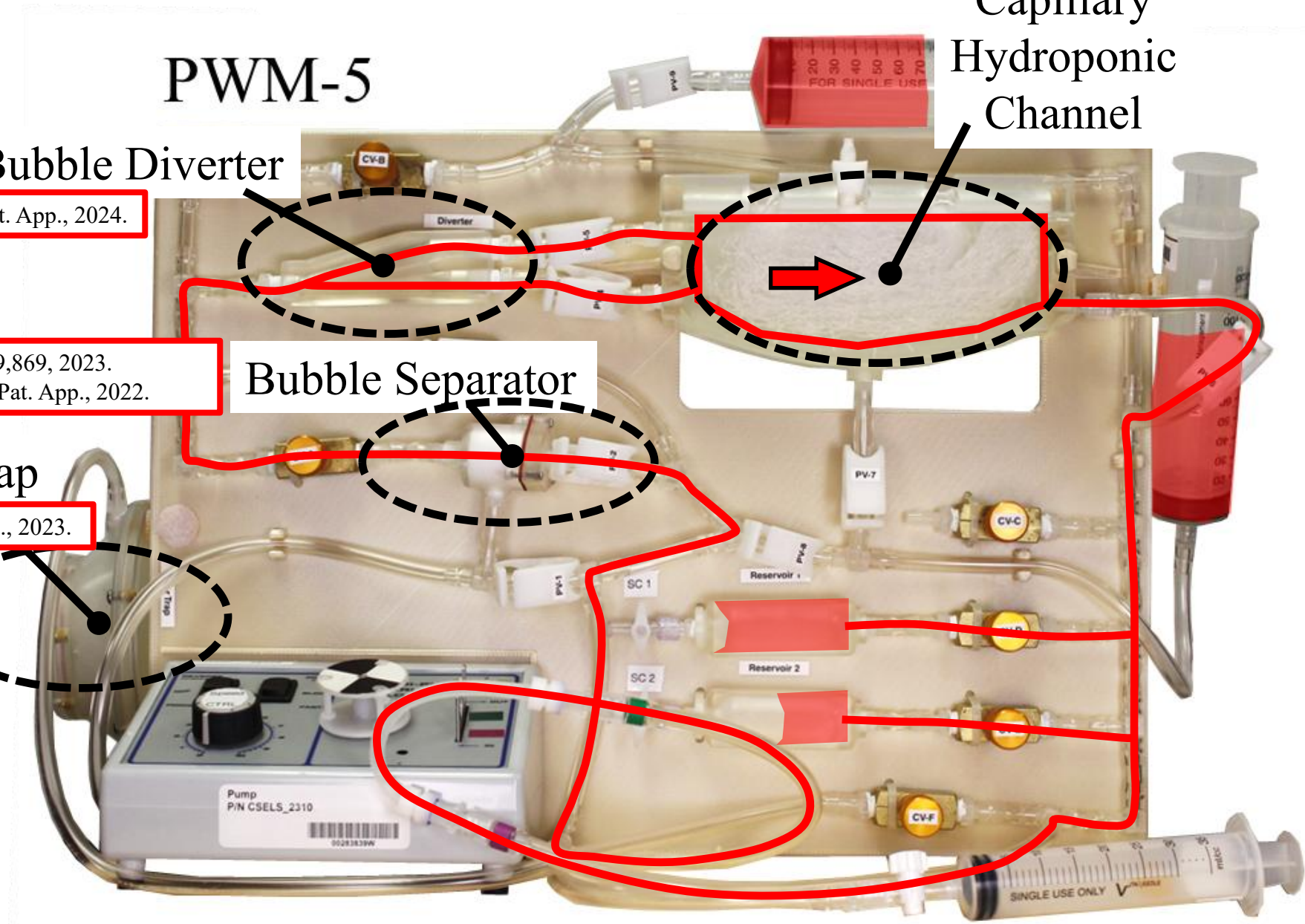
Capillary
Hydroponic
Channel

- Weislogel et al., Inertial Filter, US Pat. No. 11,779,869, 2023.
- Weislogel et al., Hydrophobic Gas Separator, US Pat. App., 2022.

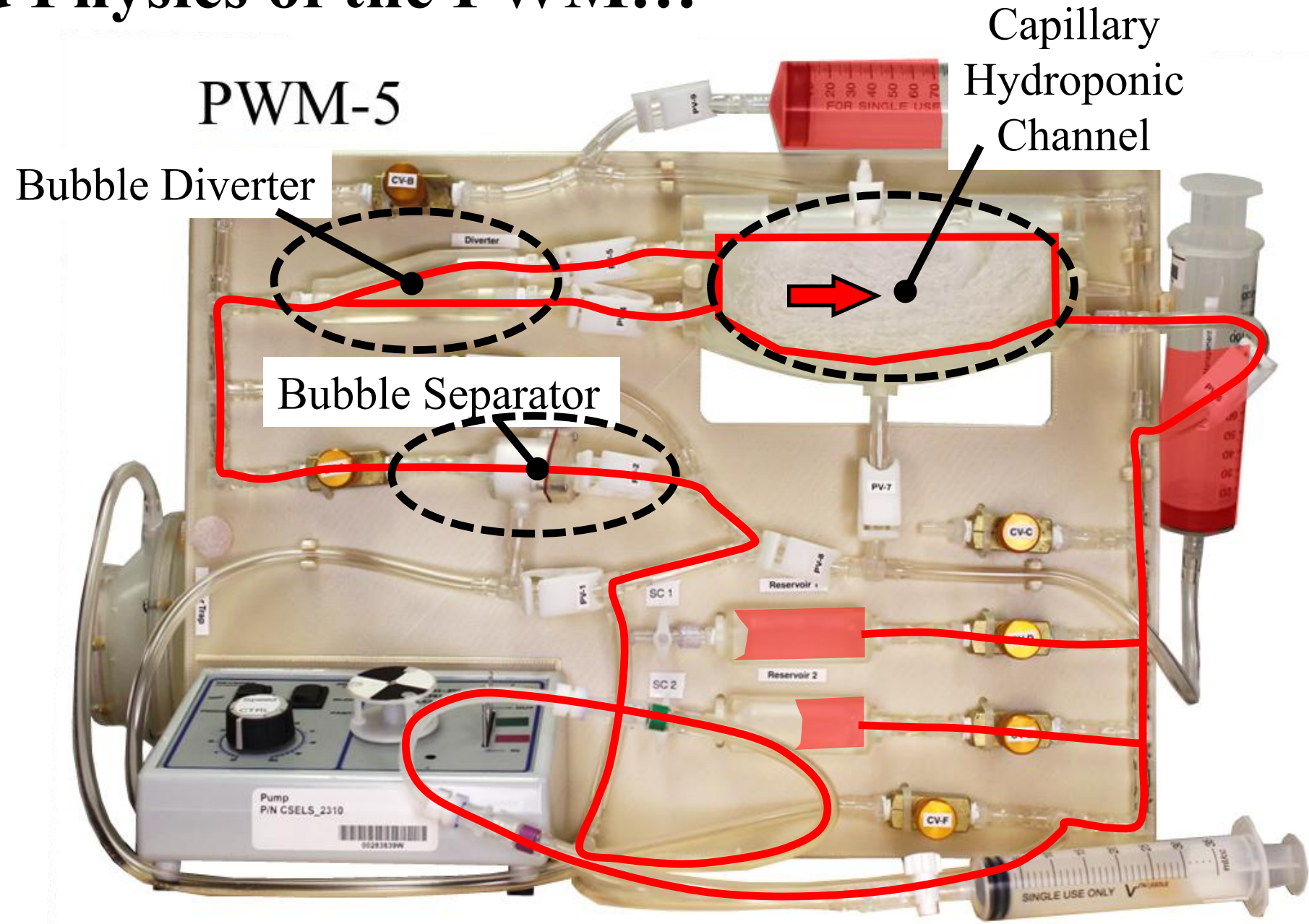
Bubble Separator

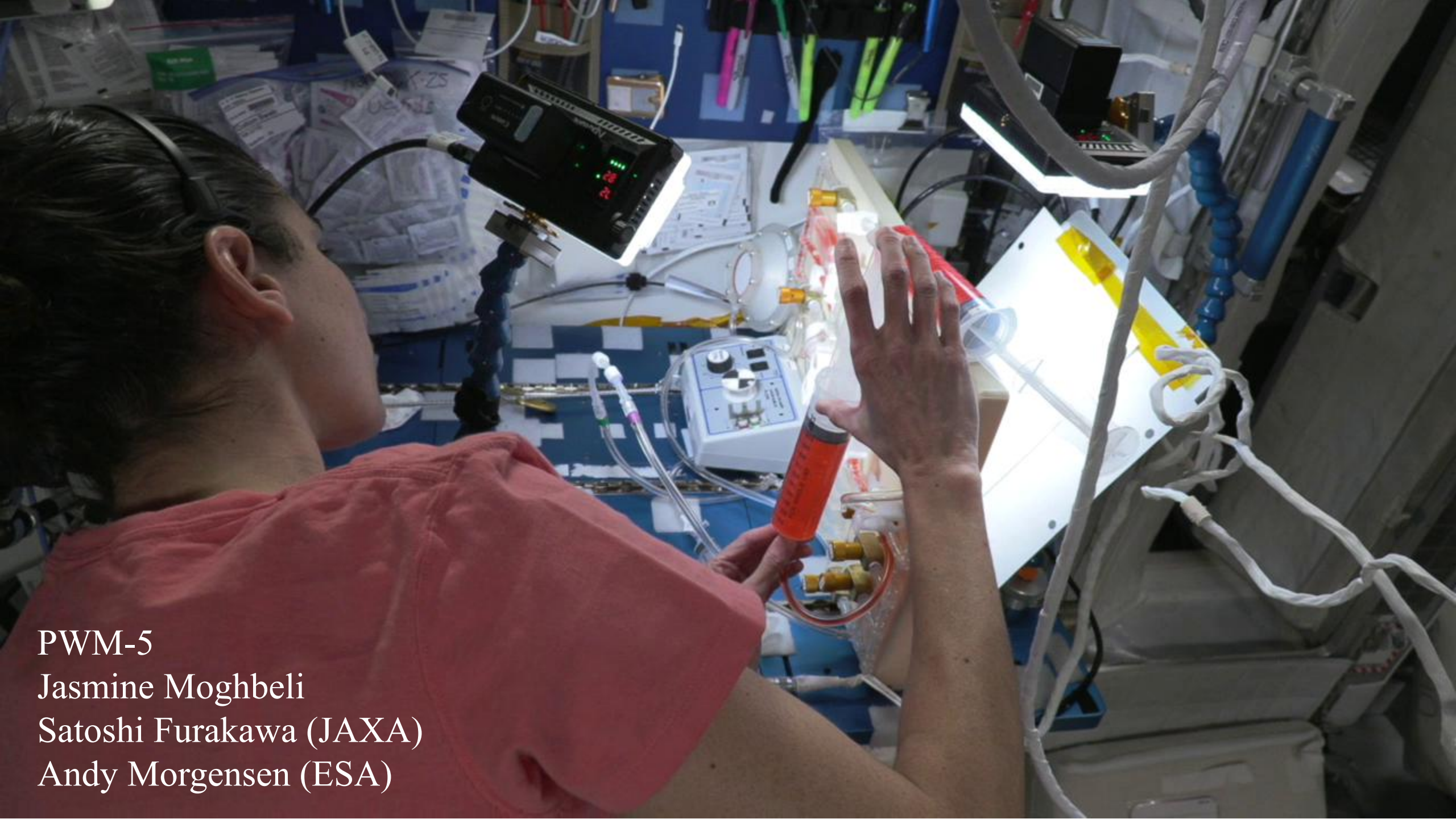
Water Trap

- Weislogel et al., Capillary Coalescing Filter, US Pat. App., 2023.



The μ -g Fluid Physics of the PWM...





PWM-5

Jasmine Moghbeli

Satoshi Furakawa (JAXA)

Andy Morgensen (ESA)



PWM-6

Suni Williams

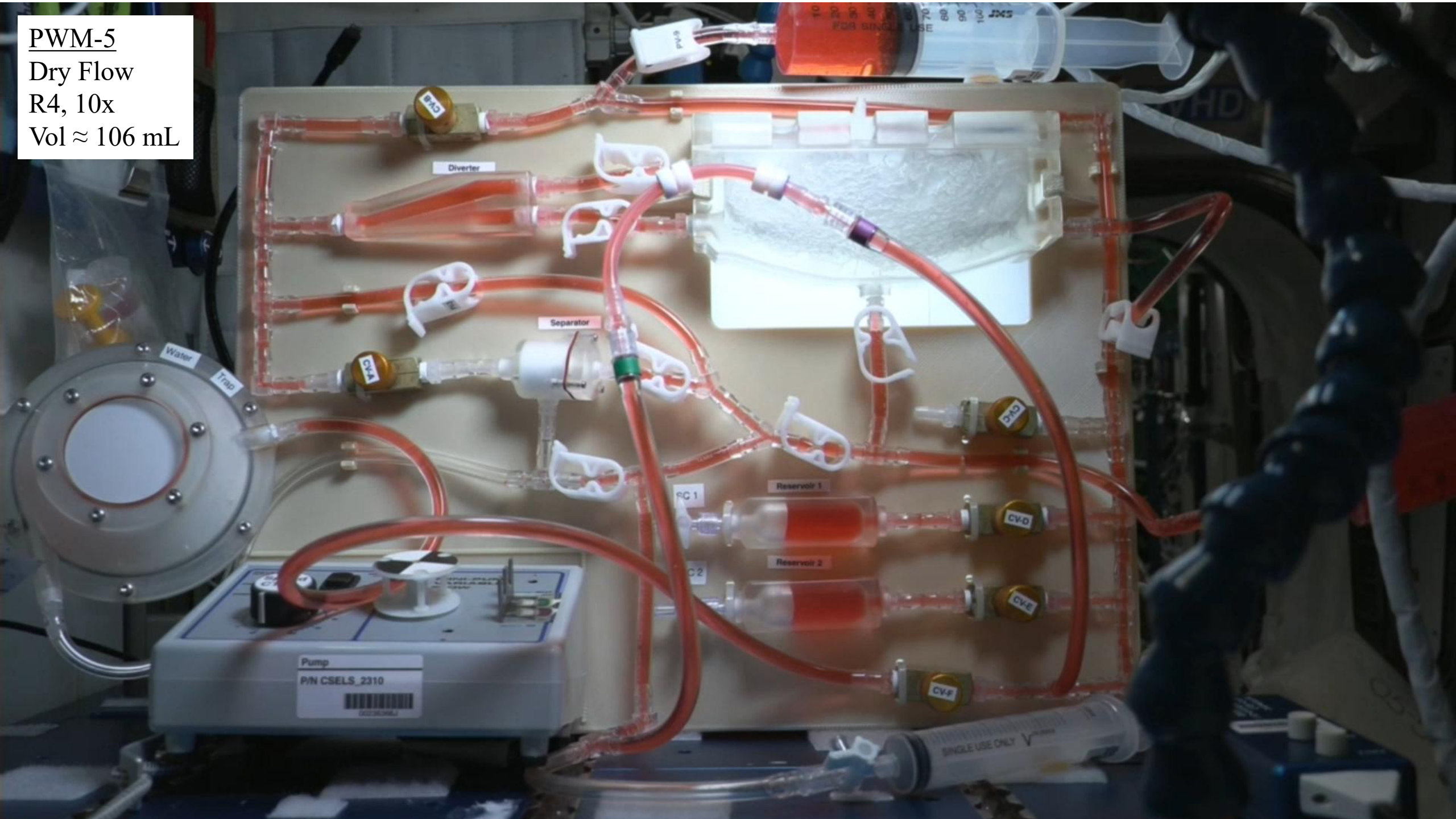


Butch Willmore



PWM-5b
Mike Barrat
Matt Dominick
Jeanette Epps

PWM-5
Dry Flow
R4, 10x
Vol $\approx$ 106 mL



E&F Summary

1. E&F has little impact on root geometry (and only slightly for sparse roots)

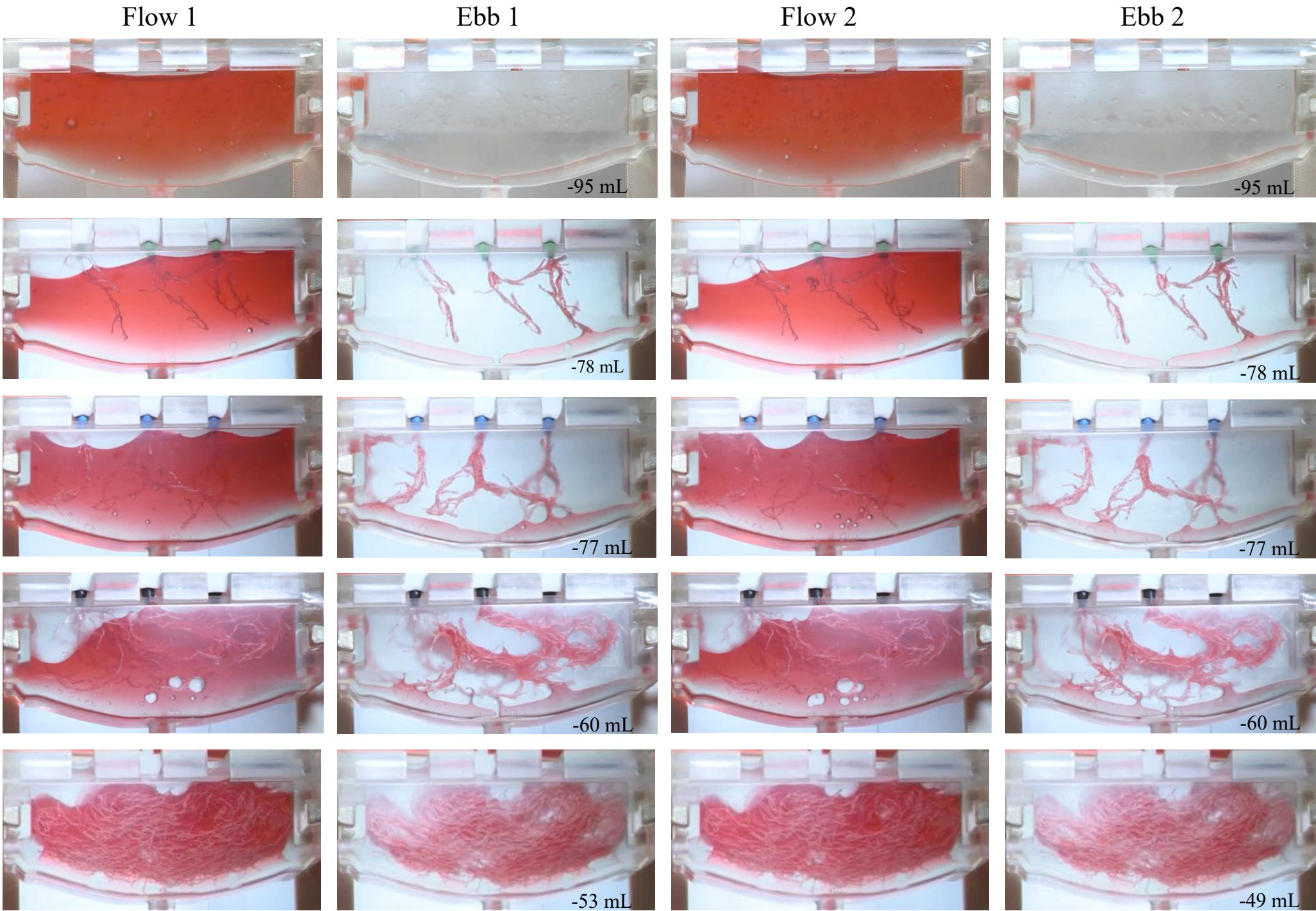
2. Root geometry has significant impact on E&F

3. E&F volume is strongly tied to channel geometry, root density, and drain location,

4. Meas. Vols. (mL)

<u>Hold-up</u>	<u>Root Vol.</u>
R0: 1	0
R1: 17	0.1
R2: 18	0.3
R3: 34	1.2
R4: 39	5.2

R0
R1
R2
R3
R4

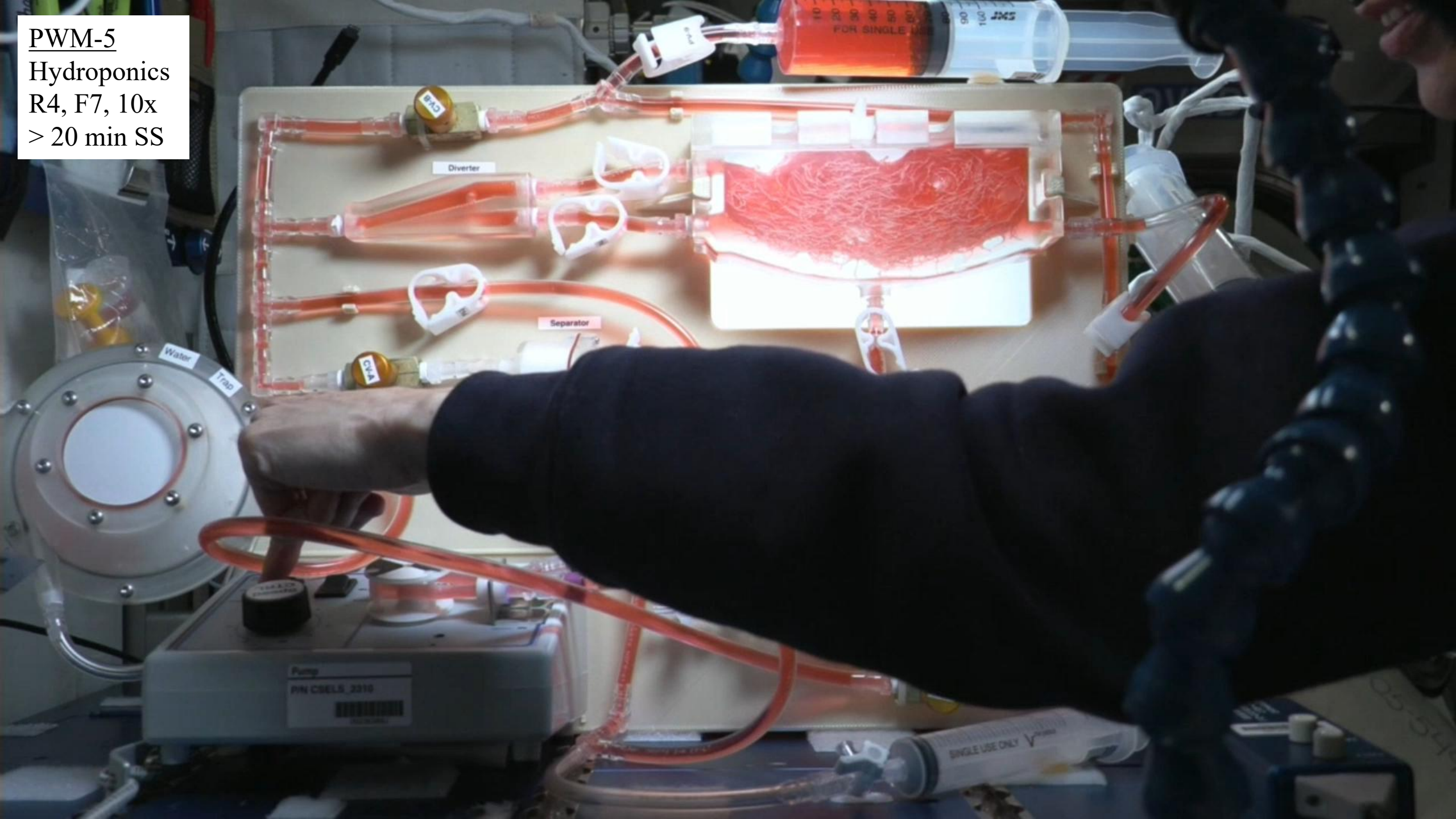


PWM-5

Hydroponics

R4, F7, 10x

> 20 min SS



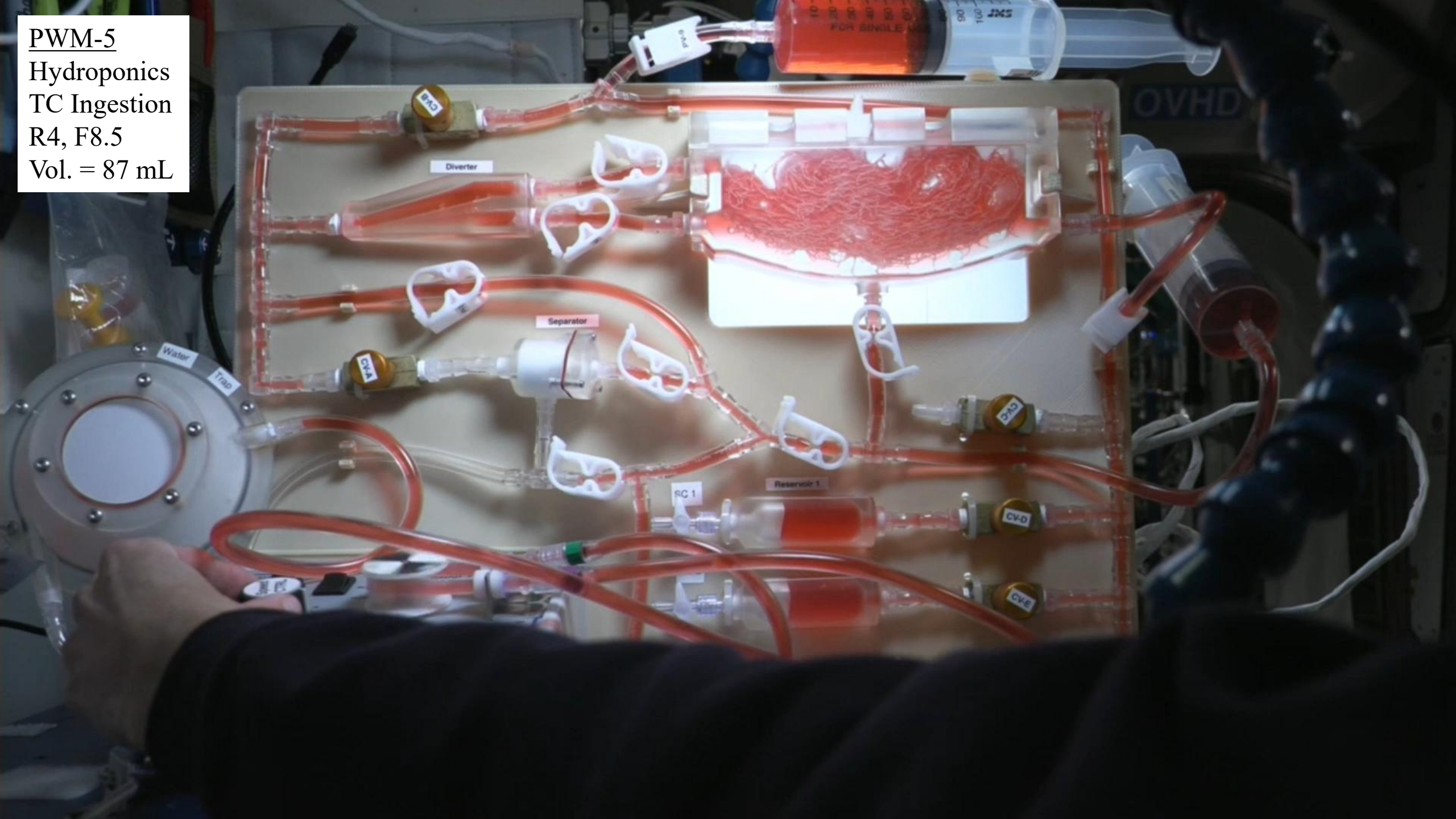
PWM-5

Hydroponics

TC Ingestion

R4, F8.5

Vol. = 87 mL



PWM-5b

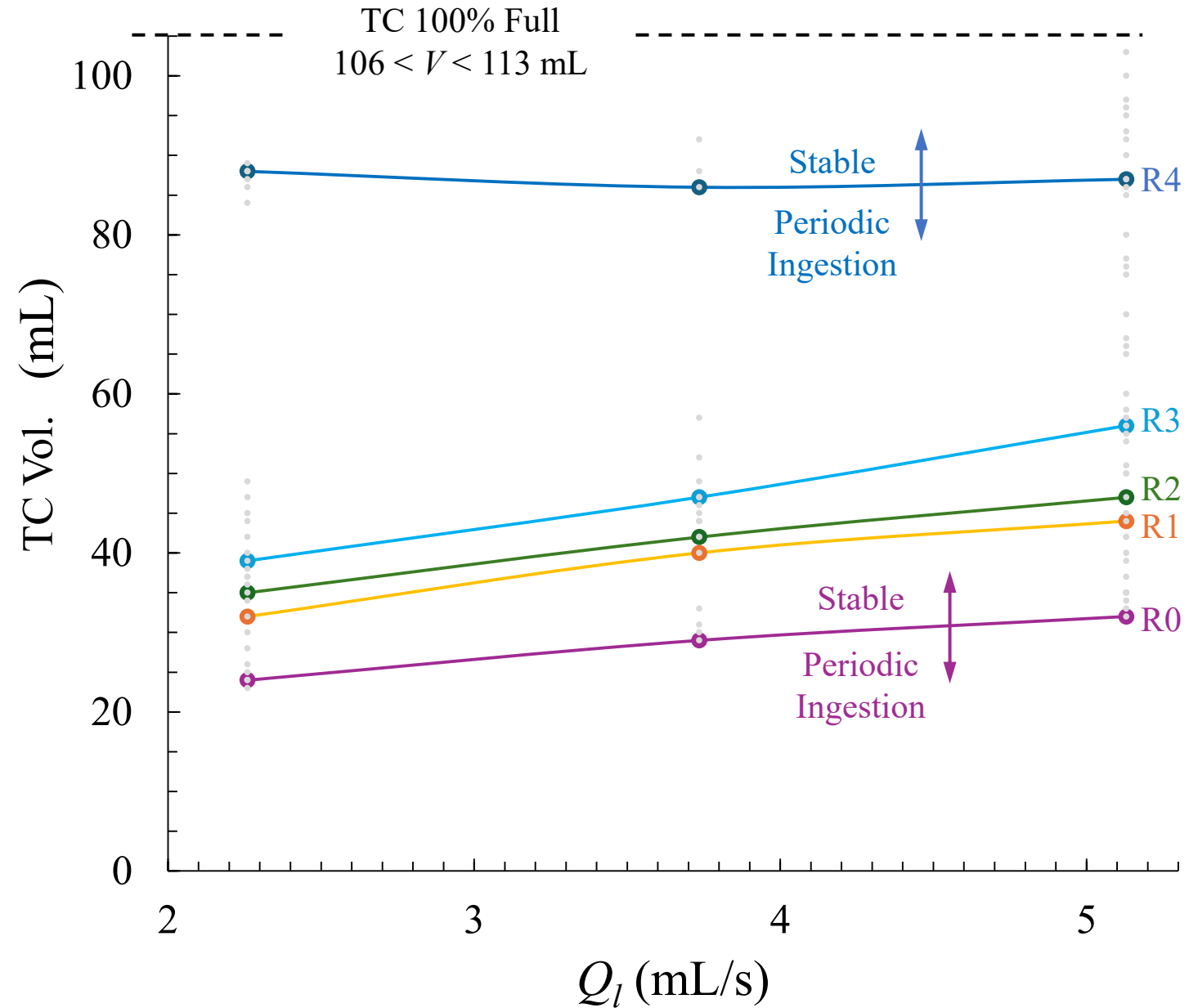
Hydroponics

TC Ingestion

Fix Q_l , reduce V

Ingestion Limit Summary

1. Sparse roots have noticeable impact on flow resistance
2. Dense roots require $\sim 3x$ more liquid for stable flow
3. Dense roots have less volume margin for stable flow



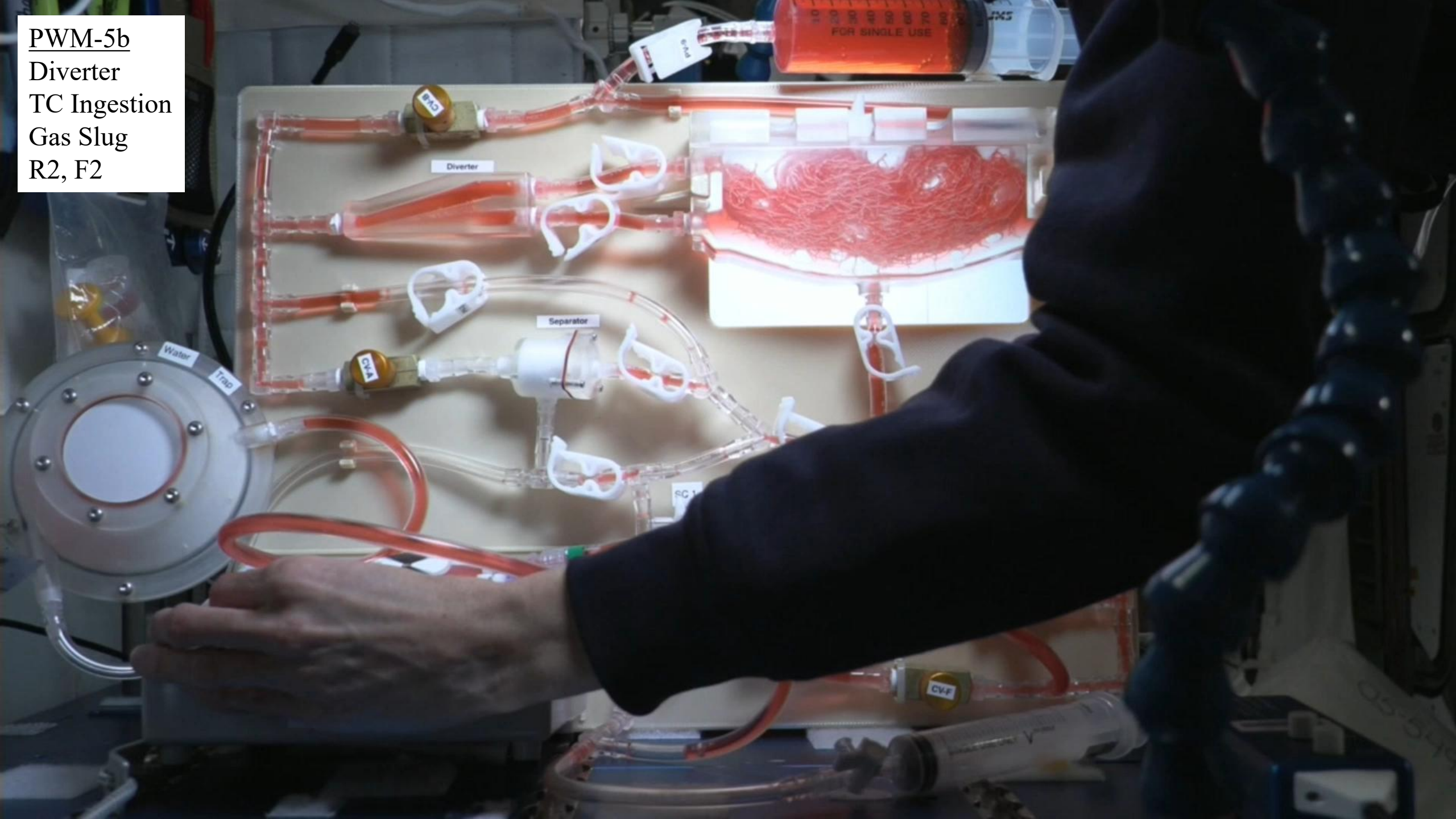
PWM-5b

Diverter

TC Ingestion

Gas Slug

R2, F2



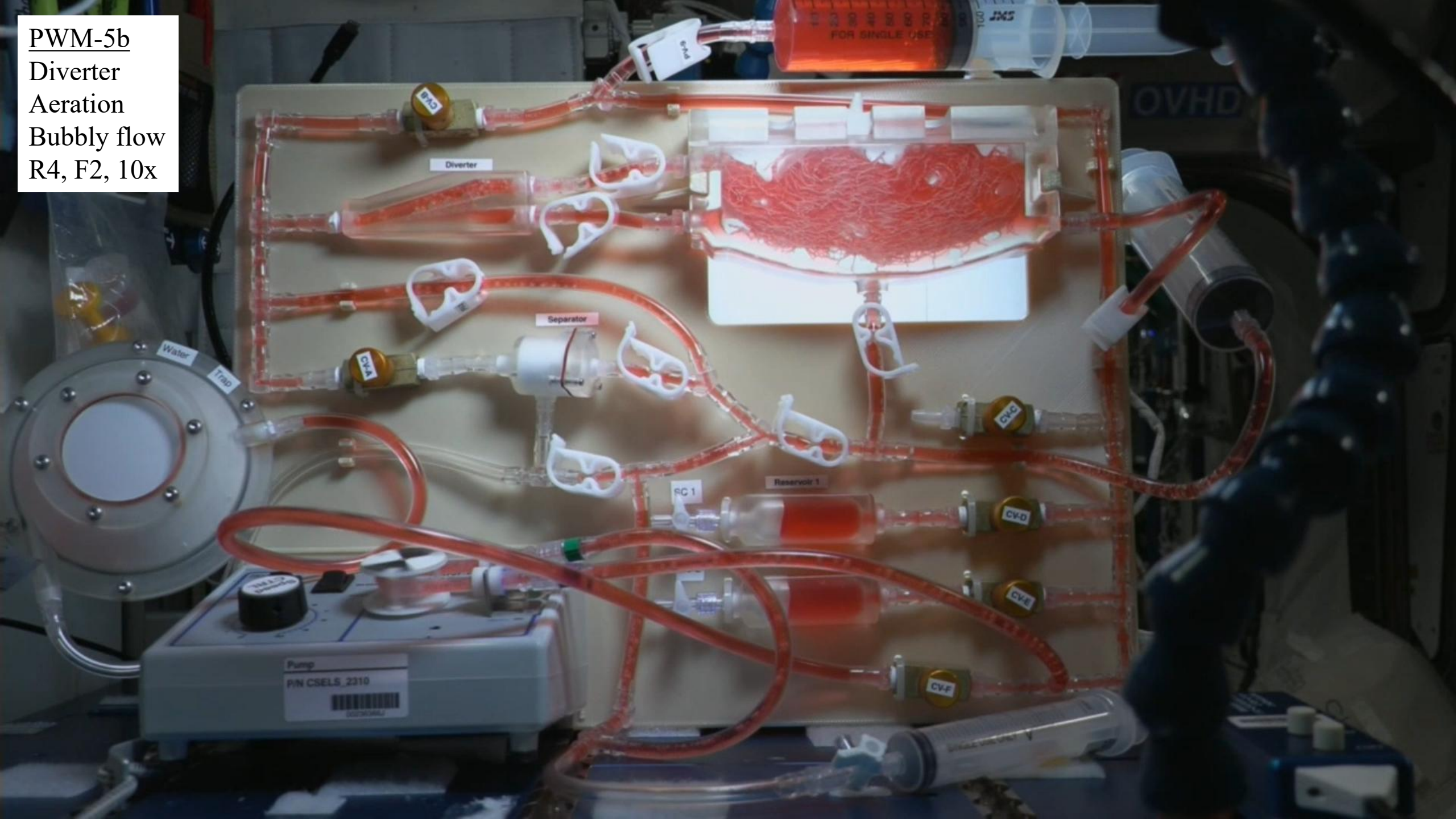
PWM-5b

Diverter

Aeration

Bubbly flow

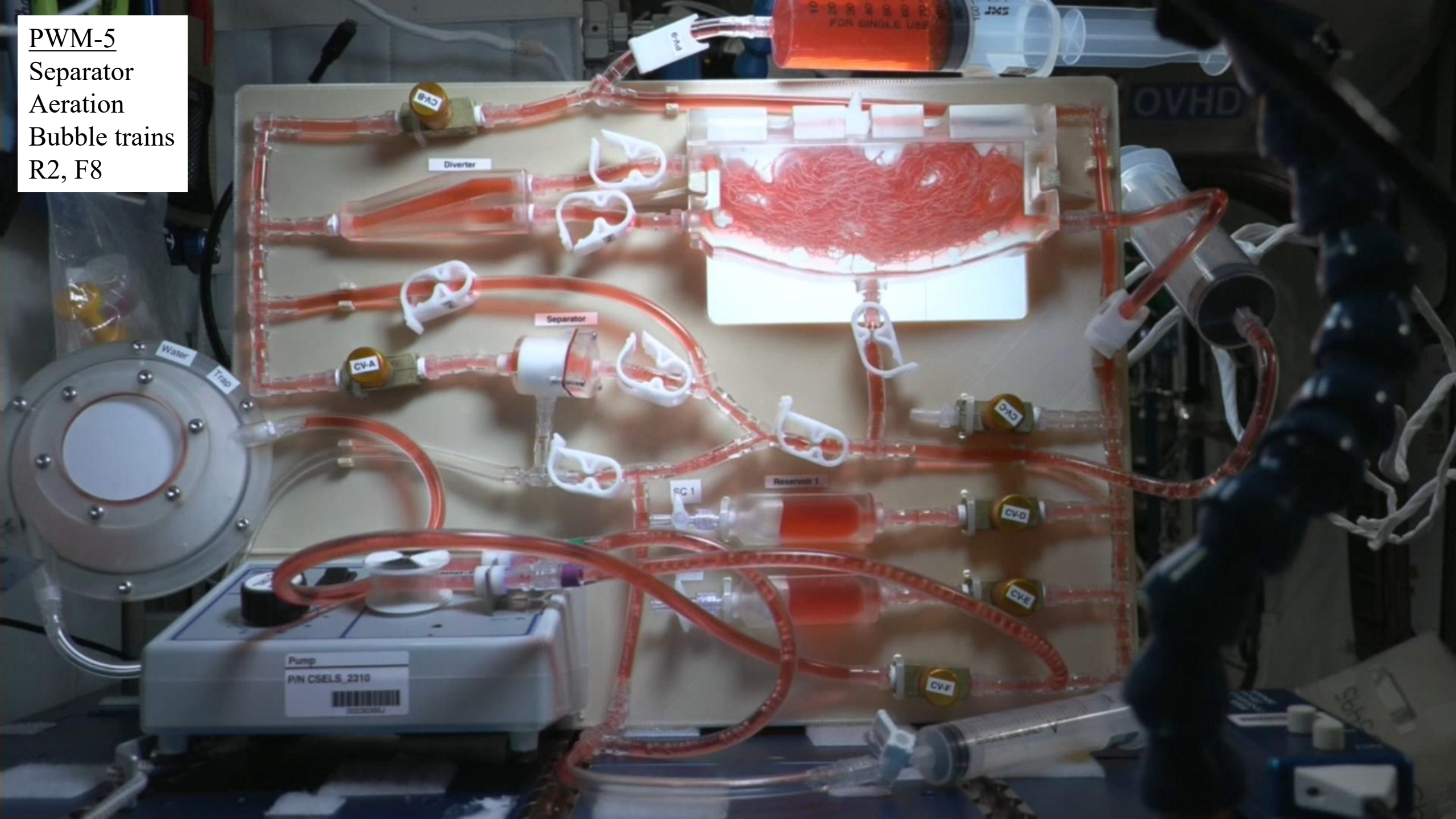
R4, F2, 10x



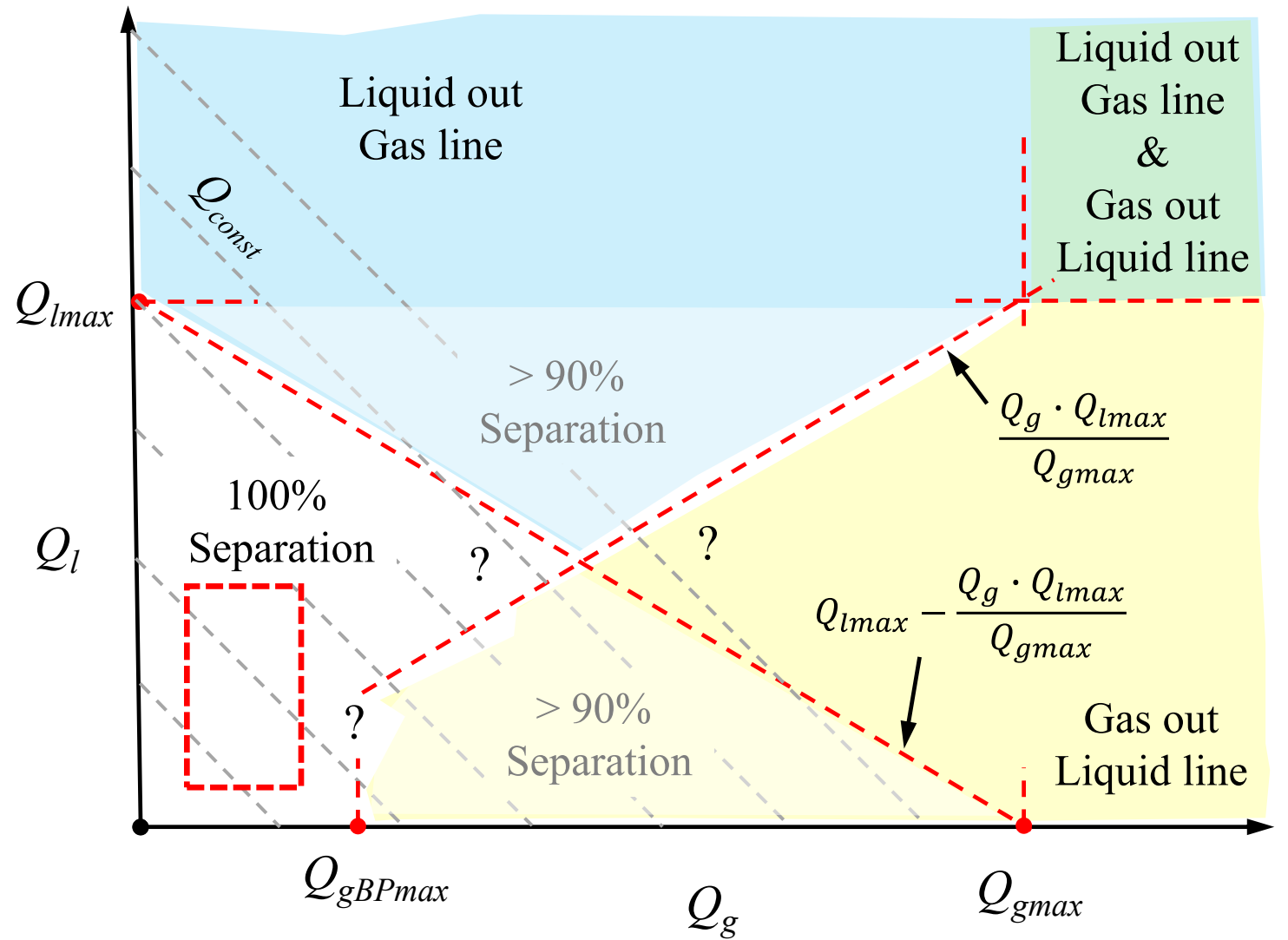
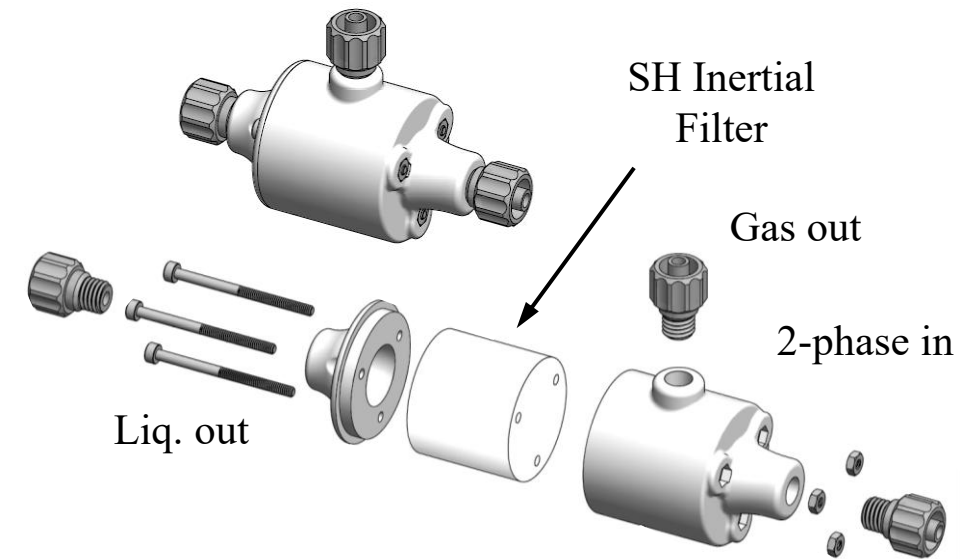
PWM-5
Diverter 100%
Aeration
R4, F5



PWM-5
Separator
Aeration
Bubble trains
R2, F8



Passive In-Line Bubble Separator



$$Q_{lmax} \sim \frac{\sigma D_c^4}{\mu_l D_p L_c}$$

$$Q_{gBPmax} \sim \frac{\epsilon \sigma D_p^2 L_c (D_c + 2\delta)}{\mu_g D_c \delta}$$

$$Q_{gmax} \sim \frac{\sigma D_c^4}{\mu_g D_p L_c}$$

	Clip	Date	GMTD	OpName	Crew	Clip Name	Clip Link	Pump Spe	Channel	Position	Wye Fitting	Bubble	Clip Description
200	199	3/31/2021	90	PWM 3/4 - Stow	Hopper	iss064m130901620_clip5	iss064m130901620_clip5	Off	Cylinder	na	Y-Diverter	No	Hardware teardown
201	200	3/31/2021	90	PWM 3/4 - Stow	Hopper	iss064m130901644_clip1	iss064m130901644_clip1	Off	Cylinder	na	Y-Diverter	No	Hardware teardown + stow
202					Mark	iss065m451461529_clip1	iss065m451461529_clip1	Off	Wedge	na	Simple Wye	No	Unwrap test stand, configure work lights, mate fill syringes
203					Mark	iss065m451461529_clip2	iss065m451461529_clip2	Off	Wedge	na	Simple Wye	No	Partial tubing prime
204					Mark	iss065m451461529_clip3	iss065m451461529_clip3	Off	Wedge	na	Simple Wye	No	Complete tubing prime, reservoir prime
205	204	5/26/2021	146	PWM 3/4 - Prime Ops	Mark	iss065m451461529_clip4	iss065m451461529_clip4	Off	Wedge	Upper	Simple Wye	No	Upper channel prime (accidental partial simultaneous prime via series bypass valve)
206	205	5/26/2021	146	PWM 3/4 - Prime Ops	Mark	iss065m451461529_clip5	iss065m451461529_clip5	Off	Wedge	Lower	Simple Wye	No	Lower channel prime, truncated by card switchover
207	206	5/26/2021	146	PWM 3/4 - Prime Ops	Mark	iss065m451461645_clip1	iss065m451461645_clip1	Off	Wedge	Lower	Simple Wye	No	Lower channel prime (continuation of iss064m451461529_clip5), Mark notes that ~375mL water added to red drink
208	207	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip2	iss065m451461645_clip2	Off	Wedge	na	Simple Wye	No	Hardware introduction
209	208	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip3	iss065m451461645_clip3	Off	Wedge	Upper	Simple Wye	No	Upper channel level adjust
210	209	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip4	iss065m451461645_clip4	Off	Wedge	Lower	Simple Wye	No	Lower channel level adjust, embolism in upper channel
211	210	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip5	iss065m451461645_clip5			Upper	Simple Wye	No	Upper channel fluid withdrawal, dry & refill to fix embolism
212	211	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip6	iss065m451461645_clip6			Lower	Simple Wye	No	Lower channel level adjust
213	212	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip7	iss065m451461645_clip7			na	Simple Wye	No	Pump intro & pump head tubing install
214	213	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip8	iss065m451461645_clip8			Upper	Simple Wye	No	Pump lowest stable speed test, pump head tubing adjustments to loosen tubing (unclear if successful)
215	214	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip9	iss065m451461645_clip9	Slow_10	Wedge	Upper	Simple Wye	No	Upper flow at lowest stable setting, hold for 3 min
216	215	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip10	iss065m451461645_clip10	Fast_2	Wedge	Upper	Simple Wye	No	Upper flow (did not run at Fast_0), hold for 3 min
217	216	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip11	iss065m451461645_clip11	Fast_2	Wedge	Upper	Simple Wye	No	Upper flow, opened red valve (upper inlet)
218	217	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip12	iss065m451461645_clip12	Fast_2	Wedge	Both	Simple Wye	No	Parallel channel flow, long-duration steady state run, Thamas description of fluid distribution in lower channel
219	218	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip13	iss065m451461645_clip13	Off	Wedge	Upper	Simple Wye	No	Upper channel fluid rebalance/withdrawal
220	219	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip14	iss065m451461645_clip14	Fast_1	Wedge	Both	Simple Wye	No	Parallel flow, long-duration steady state run, slow fluid sh to upper channel
221	220	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip15	iss065m451461645_clip15	Fast_1	Wedge	Both	Simple Wye	No	Parallel flow, engage dark blue valve 1-click, long-duration steady state run
222	221	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip16	iss065m451461645_clip16	Off	Wedge	Both	Simple Wye	No	Reconfigure for serial channel flow
223	222	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461645_clip17	iss065m451461645_clip17	Slow_7	Wedge	Both	Simple Wye	No	Serial channel flow, brief run, stop to drain liquid from upper channel to below pinning edge, truncated by card
224	223	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m141461401_clip1	iss065m141461401_clip1	Off	Wedge	Both	Simple Wye	No	Rebalance channel fill levels for serial flow, continued from iss065m451461645_clip17
225	224	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m141461401_clip2	iss065m141461401_clip2	Slow_7	Wedge	Both	Simple Wye	No	Serial channel flow, brief run, upper channel overflows
226	225	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m141461401_clip3	iss065m141461401_clip3	Off	Wedge	Both	Simple Wye	No	Rebalance channel flow, drain back fluid in upper channel
227	226	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461939_clip1	iss065m451461939_clip1	Off	Wedge	Upper	Simple Wye	No	Install 1 NR plant into upper channel, refill channel, configure for upper flow
228	227	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461939_clip2	iss065m451461939_clip2	Slow_7	Wedge	Upper	Simple Wye	No	Upper flow, 1 NR plant, hold for 10 min, stead-state run
229	228	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461939_clip3	iss065m451461939_clip3	Slow_8-->10	Wedge	Upper	Simple Wye	No	Upper flow, 1 NR plant, increase speed, hold for 10-20sec at each speed up to Slow_10
230	229	5/26/2021	146	PWM 3/4 - TST Cell1 Ops	Thomas	iss065m451461939_clip4	iss065m451461939_clip4	Fast_0-->Fast_10	Wedge	Upper	Simple Wye	No	Upper flow, 1 NR plant, hold for short duration at each speed then increase

PWM-5, -6, & -5b Summary of Accomplishments:

- 8 Crew, ~57 hr: NASA Moghbeli, Barrat, Dominick, Epps, Boeing Williams, Wilmore, JAXA Furukawa, ESA Mogensen
- ~ 4,500,000 data pts tracked
- ~ 2,000 crew/hardware interactions (pump, valves, syringes, ...)
- ~ 200 Ebb/Flow tests (#0-#4): surprising repeatability demonstrated
- ~ 120 Hydroponics tests (no bubbles) (#0-#4): ~ 20 min steady states
- ~ 100 Hydroponics tests with bubble aeration, diversion, separation, and water trap
- ~ 200 Bubble Diversion tests (#0-#4): many 100% diversions
- ~ 100 Bubbles Separation tests (#0, #1, #4): many 100% separations
- Limits of Operation measured:
 1. Test Channel: gas ingestion, fill volume, root size, bubble separations (many 100% separations)
 2. Bubble Diverter: flow rate, flow rate ratio, inlet flow regime (many 100% separations)
 3. Bubble Separator: flow rate, flow rate ratio, aeration/air injection, ΔP (many 100% separations)
 4. Water Trap: fill volume, drain rate, separation (100% trap demonstrated)
- ~ 1,000,000 bubbles generated, flowed, coalesced, and separated
- Hydroponic Modes Demonstrated: ‘Cavalier’ prime, ‘Deep’ steady states (30 min), serial/parallel flow, periodic, aperiodic, unstable, and recovery from unstable
- Dual passive reservoir demonstration
- Easy start-up, shut-down, plant changeout, ...
- **Foundations of PWM is SMD-supported Fluid Physics Research**
- **4 device patents connected to and arising from PWM technology development**
- **All devices have found cross-cutting gov’t & commercial aerospace application and sales:**
 - ECLSS: Urine processors, pretreat sensor, CHXs, ...**
- **General approach under continued development at NASA KSC**